SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Deep Learning COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

Duration: 1 Hour

QUESTIONS: 5 Total Marks:50

Annexure A

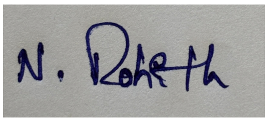
Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:
   H H T H H H T H H T
   Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).
2. Out of 50 students, 42 passed an exam.
   Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.
3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
   Find the MLE for the probability that a bulb is defective.
4. A biased die is rolled 60 times. The number 6 appears 18 times.
   Estimate the probability of rolling a 6 using MLE.
5. Given:
   Input (x) = 4
   Weight (w) = 2
   Actual Output = 10
   Prediction:
   $y = w \times x$
   1. Find the predicted output.
   2. Calculate the error.
   3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Code of Conduct:

I ***<u>NALIPI REDDY ROHITH/192425115</u>****(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Analytical Problem Solving

Q1. A coin is tossed 10 times and the outcomes are:

H H T H H H T H H T

Question:

Using **Maximum Likelihood Estimation (MLE)**, estimate the probability of getting Heads ((p)).

Solution:

Step 1: Count the number of Heads

Heads = **7**

Tails = **3**

Total Tosses = **10**

Step 2: Apply the MLE Formula

For a Bernoulli experiment, the Maximum Likelihood Estimator (MLE) of the probability of success is

[
\hat{p}=\frac{\text{Number of Successes}}{\text{Total Number of Trials}}
]

Substituting the given values,

[
\hat{p}=\frac{7}{10}=0.7
]

Interpretation

This means the estimated probability of obtaining a Head is **0.7**, or **70%**. Based on the observed data, the coin appears to favor Heads.

Conclusion

The MLE estimate of the probability of getting a Head is

[
\boxed{\hat{p}=0.7}
]

Answer: 0.7 (70%)

Q2. Out of 50 students, 42 passed an exam.

Question:

Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

Solution:

Each student's result can be considered a **Bernoulli trial**, where:

- Pass = Success
- Fail = Failure

Step 1: Count the successes

Number of students passed = **42**

Total students = **50**

Step 2: Apply the MLE Formula

$$
\hat{p}=\frac{\text{Number of Passes}}{\text{Total Students}}
$$

$$
\hat{p}=\frac{42}{50}=0.84
$$

Interpretation

The estimated probability that a randomly selected student passes the exam is **84%**.

Conclusion

The MLE estimate for the probability of passing is

$$
\boxed{\hat{p}=0.84}
$$

Answer: 0.84 (84%)

Q3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.

Question:

Find the MLE for the probability that a bulb is defective.

Solution:

A bulb is either **defective** or **non-defective**, so the data follows a **Bernoulli distribution**.

Step 1: Identify the data

Defective bulbs = **8**

Total bulbs = **100**

Step 2: Apply the MLE Formula

$$
\hat{p}=\frac{\text{Defective Bulbs}}{\text{Total Bulbs}}
$$

$$
\hat{p}=\frac{8}{100}=0.08
$$

Interpretation

The estimated probability that a bulb is defective is **0.08**, which means **8%** of the bulbs are expected to be defective.

Conclusion

The MLE estimate of the defect probability is

$$
\boxed{\hat{p}=0.08}
$$

Answer: 0.08 (8%)

Q4. A biased die is rolled 60 times. The number 6 appears 18 times.

Question:

Estimate the probability of rolling a 6 using Maximum Likelihood Estimation (MLE).

Solution:

Rolling a **6** is treated as a success.

Step 1: Identify the observations

Number of sixes = **18**

Total rolls = **60**

Step 2: Apply the MLE Formula

[
\hat{p}=\frac{\text{Number of Sixes}}{\text{Total Rolls}}
]

[
\hat{p}=\frac{18}{60}=0.30
]

Interpretation

The estimated probability of rolling a 6 is **0.30**, or **30%**. Since a fair die has a probability of **1/6 ≈ 0.167**, this die appears to be biased toward rolling a 6 more frequently.

Conclusion

The MLE estimate is

[
\boxed{\hat{p}=0.30}
]

Answer: 0.30 (30%)

Q5. Given:

- Input ((x)) = **4**
- Weight ((w)) = **2**
- Actual Output = **10**

Prediction Formula:

[
y=w\times x
]

(a) Find the Predicted Output

Using the prediction equation,

[
y=2\times4
]

[
y=8
]

Predicted Output = 8

(b) Calculate the Error

Error is calculated as

[
\text{Error}=\text{Actual Output}-\text{Predicted Output}
]

Substituting the values,

[
=10-8
]

[
=2
]

Error = 2

(c) Update the Weight

Given:

- Current Weight = **2**

- Gradient = **4**
- Learning Rate = **0.01**

The Gradient Descent weight update rule is

[
w_{\text{new}}=w-\eta \times \text{gradient}
]

Substituting the values,

[
w_{\text{new}}=2-(0.01\times4)
]

[
=2-0.04
]

[
=1.96
]

Interpretation

Since the prediction error is positive, the weight is slightly reduced to minimize the loss in the next iteration. Gradient Descent repeatedly updates the weight until the prediction error becomes very small.

Conclusion

- **Predicted Output = 8**
- **Error = 2**
- **Updated Weight = 1.96**

[
\boxed{w_{\text{new}}=1.96}
]

These expanded solutions are suitable for **20-mark university exam answers**, with step-by-step calculations, formulas, interpretation, and conclusion for each problem.